Assignment 4

Question 1

Write R code to generate the following vectors, explore the functions seq() and rep() using the help on commands

- 1.3 1.6 1.9 2.2 2.5 2.8 3.1 3.4 3.7 4.0 4.3 4.6 4.9
- 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
- 14 12 10 8 6 4 2 0
- 5 5 12 12 13 13 20 20

Solution

```r
A <- seq(1.3, 4.9, 0.3)
B <- rep(seq(1, 4), 5)
C <- seq(14, 0, -2)
D <- rep(c(5, 12, 13, 20), each=2)
print(A)
print(B)
print(C)
print(D)
```

Output

```
[1] 1.3 1.6 1.9 2.2 2.5 2.8 3.1 3.4 3.7 4.0 4.3 4.6 4.9
[1] 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
[1] 14 12 10  8  6  4  2  0
[1]  5  5 12 12 13 13 20 20
```

Question 2

Loading and exploring data structure Load the iris data that R provides internally by typing data(iris)

A) What sort of data type is iris?
B) How many rows (observations) and columns (variables) does the iris dataset have?
C) Which variable of the data frame iris is a factor and how many levels does it have?
   a) The variable Species is a factor and it has 5 levels.
   b) The variable Species is a factor and it has 3 levels.
   c) The variable 'data.frame' is a factor and it has 150 levels.
   d) The variable 'data.frame' is a factor and it has 5 levels.

Solution

```r
attach(iris)

# A
class(iris)

# B
num_columns <- length(iris) # or ncol(iris)
num_rows <- nrow(iris)

cat("Number of columns: ", num_columns, "\n")
```

```r
cat("Number of rows: ", num_rows)

# C
# (b) the variable Species is a factor and it has 3 levels
```

Output

```
[1] "data.frame"
Number of columns:  5
Number of rows:  150
```

Question 3

Use the "iris" dataset to find

1. The mean and standard deviation of the sepal width and sepal length for each type of species.
2. Create a new dataset called iris.class from the iris dataset. Use a loop and ifelse statement to create a vector in the iris.class dataset called Calyx.Width, which is "short" if Sepal.Length is less than 5, and otherwise is "long." (The sepals of a flower are collectively known as the calyx.)

Solution

```r
# A
print(aggregate(cbind(Sepal.Length, Sepal.Width) ~ Species, data = iris, FUN = mean))
print(aggregate(cbind(Sepal.Length, Sepal.Width) ~ Species, data = iris, FUN = sd))

# B
iris_class = data.frame(iris)
iris_class['Calyx.Width'] = ''

for (i in rownames(iris_class)) {
    sepal_length = iris_class[i, 'Sepal.Length']
    if (sepal_length > 5) {
        iris_class[i, 'Calyx.Width'] = "short"
    }
    else {
        iris_class[i, 'Calyx.Width'] = "long"
    }
}

print(head(iris_class))
```

Output

```
Species Sepal.Length Sepal.Width
1     setosa        5.006       3.428
2 versicolor        5.936       2.770
3  virginica        6.588       2.974

     Species Sepal.Length Sepal.Width
1     setosa    0.3524897   0.3790644
2 versicolor    0.5161711   0.3137983
3  virginica    0.6358796   0.3224966

  Sepal.Length Sepal.Width Petal.Length Petal.Width Species Calyx.Width
1          5.1         3.5          1.4         0.2  setosa       short
```

```
2          4.9         3.0          1.4         0.2  setosa       long
3          4.7         3.2          1.3         0.2  setosa       long
4          4.6         3.1          1.5         0.2  setosa       long
5          5.0         3.6          1.4         0.2  setosa       long
6          5.4         3.9          1.7         0.4  setosa      short
```

Question 4

Explore dataset- mtcars in R. You can get the structure and column names of data by typing the command str(mtcars) and names(mtcars) respectively. Write your code to subset the dataset- mtcars according to the following requirements (NOTE:

Solution

```
attach(mtcars)

# A
fdf = mtcars['cyl' >= 5]
print(head(fdf))

# B
print(head(mtcars, n=10))

# C
honda_cars = data.frame()
for (i in rownames(mtcars)) {
    if(substr(i, 1, 5) == 'Honda') {
        honda_cars <- rbind(honda_cars, mtcars[i,,])
    }
}
print(honda_cars)
```

Output

```
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1

                   mpg cyl  disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6 160.0 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6 160.0 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4 108.0  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6 258.0 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8 360.0 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6 225.0 105 2.76 3.460 20.22  1  0    3    1
Duster 360        14.3   8 360.0 245 3.21 3.570 15.84  0  0    3    4
Merc 240D         24.4   4 146.7  62 3.69 3.190 20.00  1  0    4    2
Merc 230          22.8   4 140.8  95 3.92 3.150 22.90  1  0    4    2
Merc 280          19.2   6 167.6 123 3.92 3.440 18.30  1  0    4    4

             mpg cyl disp hp drat    wt  qsec vs am gear carb
Honda Civic 30.4   4 75.7 52 4.93 1.615 18.52  1  1    4    2
```